IJIRSET
International Journal of Innovative Research in
SCIENCE | ENGINEERING | TECHNOLOGY

e-ISSN: 2319-8753 | p-ISSN: 2347-6710

INTERNATIONAL JOURNAL OF INNOVATIVE RESEARCH

IN SCIENCE | ENGINEERING | TECHNOLOGY

Volume 11, Issue 9, September 2022

INTERNATIONAL STANDARD SERIAL NUMBER INDIA

Impact Factor: 8.118

9940 572 462 | 6381 907 438 | ijirset@gmail.com | www.ijirset.com

International Journal of Innovative Research in Science, Engineering and Technology (IJIRSET)

|e-ISSN: 2319-8753, p-ISSN: 2347-6710| www.ijirset.com | Impact Factor: 8.118|

|| Volume 11, Issue 9, September 2022 ||

| DOI:10.15680/IJIRSET.2022.1109004 |

Quantum Machine Learning: Harnessing Quantum Algorithms for Supervised and Unsupervised Learning

Mohit Mittal

Dr. A.P.J. Abdul Kalam Technical University, Lucknow, Uttar Pradesh, India

ABSTRACT: Quantum machine learning (QML) provides a transformative approach to data analysis by integrating the principles of quantum computing with classical machine learning methods. With the exponential growth of data and the increasing complexity of computational tasks, quantum algorithms offer tremendous advantages in terms of processing speed, memory efficiency, and the ability to resolve issues intractable for classical systems. In this work, the use of QML techniques for both supervised and unsupervised learning problems is explored. Quantum-enhanced models such Quantum Support Vector Machines (QSVMs) and Quantum Neural Networks (QNNs) show outstanding performance in classification and regression tasks by using quantum kernels and entanglement in supervised learning. Moreover, hybrid quantum-classical solutions offer useful implementations on noisy intermediate-scale quantum (NISQ) devices, hence bridging the gap between present quantum technology and practical uses. By means of comparative analysis, this paper emphasizes the possible benefits and drawbacks of QML, thereby providing understanding of its future importance in sectors including material science, finance, and healthcare. In the end, QML opens the path for a new era of intelligent data processing and solves until unthinkable difficult challenges.

KEYWORDS: Quantum Machine Learning, QSVM, Quantum Kernel Methods, IBM Qiskit, Execution Time, Accuracy

I. INTRODUCTION

Gordon E. Moore, a co-founder of Intel Inc., predicted in 1965 that "the number of transistors incorporated in a chip will approximately double every 24 months" [1]. The difficulty of producing miniaturised transistors at an economical price has resulted in a deceleration of Moore's law in recent years. The likelihood of electrical leakage escalates as transistor dimensions diminish, and processors generate heat as well. The expenses for cooling the computing systems are increasing due to the heat. As transistors diminish to microscopic dimensions, materials demonstrate quantum mechanical properties [2]. In response to the emergence of quantum computing, companies such as IBM and Google commenced the production of quantum processors utilising materials that exhibit quantum mechanical properties. Quantum computers, owing to their fundamentally distinct operation, are adept at addressing complex optimisation problems that conventional computers are incapable of handling. The exponential scalability provided by quantum computers renders them appropriate for managing extensive data. Consequently, machine learning algorithms can gain from improvements in quantum computing [3].

A quantum computer is an extremely complicated computing system that operates according to the rules of quantum physics, quantum information theory, and computer science. There are many key ways in which quantum computers vary from their conventional counterparts. A classical computer's inner workings may be better understood with the help of quantum physics, which provides the groundwork for the physical world. On the other hand, conventional computers do not handle data using quantum mechanical features. When asked about the need of quantum mechanical processing systems for modeling nature, Richard P. Feynman said in a 1981 physics lecture, "Nature is not classical." For computations, quantum computers make use of entanglement and superposition, two particular quantum mechanical features [4].

A quantum computer's processing power grows at an exponential rate. The ability to define more linear combinations with more qubits grows as the number of qubits increases. A classical computer's processing capability grows in a straight line as the number of transistors per square inch on a chip does. Specialized hardware that processes data using qubits is necessary for quantum computers. Quantum bits (qubits) are created by measuring subatomic particles, which

|e-ISSN: 2319-8753, p-ISSN: 2347-6710| www.ijirset.com | Impact Factor: 8.118|

|| Volume 11, Issue 9, September 2022 ||

| DOI:10.15680/IJIRSET.2022.1109004 |

might be in one of two distinct states. To the contrary, complementary metal-oxide semiconductor (CMOS) circuitry is the backbone of classical computers [5].

Because of the powerful quantum tools for linear algebra, quantum computing may improve conventional machine learning methods [6]. Machine learning systems may take use of quantum computing techniques, which outperform conventional approaches in real applications, since machine learning is based on linear algebra [7]. Improving machine learning algorithms is similar to manipulating qubit states by randomly adjusting gate settings to get a certain outcome. To solve problems with machine learning, the QML approach may be run as a quantum circuit using a series of quantum gate operations. Make better use of large data processing for important, practical applications by using quantum machine learning. The amount of qubits that quantum computers can have is currently limited. While there are many benefits to using quantum computers, current versions are limited by noise. Because of their interactions with their surroundings, qubits pose a threat to the information they hold. Therefore, it is difficult to solve machine learning problems with modern noisy intermediate-scale quantum devices (NISQs). In order to make sure that QML algorithms work with modern quantum computers, their designers need to think about their limitations [8].

II. RELATED WORK

The relevant literature on classical-hybrid algorithms for pattern classification is reviewed. For the purpose of pattern recognition and localization in images, Schützhold developed a quantum Fourier transform classifier. When compared to its traditional counterpart, his suggested research showed an exponential acceleration. Classifiers using adiabatic quantum computing considerably outperformed the traditional machine learning classifier AdaBoost, according to Neven, Denchev, Rose, and Macready. A quantum machine learning approach was suggested by Sentís, Calsamiglia, Muñoz-Tapia, and Bagan for the binary classification of qubit states. This technique does not rely on quantum memory and can achieve the least allowable error rate regardless of the size of the training dataset [9].

Lloyd et al. proposed a novel quantum adiabatic method variant for M vectors clustering into k groups. The amount of vectors and their dimensions may be increased exponentially using this method, which can be completed in logarithmic time [10]. In addition, they demonstrated that QML helped enhance privacy by decreasing the amount of queries to the quantum database while cluster assignment was being done. With logarithmic time complexity and exponential improvements in the size and number of training vectors, Rebentrost, Mohseni, and Lloyd have presented a quantum version of the popular support vector machine classifier, demonstrating its performance [11].

Using a distance-based classification approach similar to the K-nearest neighbour (kNN) technique, Schuld, Sinayskiy, Petruccione, and Pham presented a quantum pattern classification algorithm [12]. In this method, we generated a superposition of the training dataset and then determined the input state-to-vector amplitude Hamming distance. A qubit for the ancilla, one for the input state, and a second for the superposition were the three quantum registers used to create the quantum circuit. Using the well-known MNIST handwritten digits database, the half-moon dataset, and disease-related datasets, Wiebe, Kapoor, and Svore demonstrated the effectiveness of their quantum nearest-neighbor algorithms. These algorithms were created using established faster quantum techniques for computing distance metrics. When contrasted with their classical equivalents, they attained accelerations that were exponential or even super-exponential. An algorithm for quantum linear regression using the least squares approach on a given dataset was proposed by the researchers. In addition to determining if the dataset in issue is appropriate for linear regression, the software ran in polylogarithmic time.

Improved accuracy on real-world datasets has been shown by new supervised quantum classification techniques. A supervised learning QNN model that can represent both classical and quantum input was introduced by Farhi and Neven. Their study was an exploratory one, providing a classical model of a small quantum system optimized for future quantum computers. Class 3 and class 6 handwritten digits were binary classified using the proposed QNN on the well-known MNIST dataset [14].

Combining dimensionality reduction with a classification approach based on quantum Frobenius distance (QFD), Kerenidis and Luongo presented a quantum version of slow feature analysis (QSFA). The results were shown on the MNIST dataset, where they achieved a 98.5% classification accuracy in polylogarithmic time [15]. An efficient universal approximator of continuous functions was suggested by Torrontegui and García-Ripoll in the form of a unitary quantum perceptron that incorporates sigmoid activation. The integration of the perceptron's response into a

|e-ISSN: 2319-8753, p-ISSN: 2347-6710| www.ijirset.com | Impact Factor: 8.118|

|| Volume 11, Issue 9, September 2022 ||

| DOI:10.15680/IJIRSET.2022.1109004 |

neural network proved that this quantum neural network can perform at least as well as conventional neural networks [16].

For dimensionality reduction and classification, Liang, S.Q. Shen, M. Li, and L. Li recently presented a quantum local discriminant embedding and a quantum neighbourhood preserving embedding. Both approaches outperformed their classical counterparts exponentially faster [17]. Quantum machine learning and quantum-like machine learning were compared in Sergioli's recent study [18].

Through the use of QML approaches, conventional machine learning algorithms have been transformed into quantum algorithms that can be executed on quantum computers. Formulating and running quantum-like algorithms on classical computers is what the classical-classical technique (CC) is all about. For a synopsis of the growing body of work in the field of QML, see the bibliometric study by Pande and Mulay [19]. A thorough comparison table of several QML algorithms for solving classification and regression problems has been attempted by Abohashima, Elhosen, Houssein, and Mohamed [20]. Their emphasis was on quantum, hybrid classical-quantum, and quantum-like implementations.
S. Lloyd et al. (2013) [21] examined both supervised and unsupervised machine learning techniques in their discourse on cluster discovery and assignment. The authors presented a quantum machine learning algorithm that assigns N-dimensional vectors to a cluster of M states in a quantum computer, requiring O(log(MN)) time. The classical version, conversely, necessitates (poly(MN)) time. This discovery indicates that QML offers a computational advantage, even when managing a substantial quantity of vectors. This approach functions as a subroutine for the K-means clustering algorithm, yielding exponential acceleration.

The complexity of a Support Vector Machine (SVM) classifier on a quantum computer is logarithmic in relation to the dimensionality of the features and the quantity of training data, as proposed by P. Rebentrost et al. [22]. Phase estimation and matrix inversion are performed using a least-squares formulation of SVM. The quantum SVM employs PCA to get enhanced performance. The authors established the theoretical foundation for QSVM.

J.C. Adcock et al. [23] presented a comparative comparison of standard machine learning (ML) and quantum machine learning (QML), highlighting the application of principal component analysis in QML. The authors discussed various quantum algorithms, including HHL, SVM, QSVM, k-Nearest Neighbour (KNN), among others. To determine if a handwritten numeral is 9 or 6, a 4-qubit quantum simulator and a QSVM-based model were employed.

M. Schuld et al. [24] conducted a thorough study of QML, including the application of quantum supervised and unsupervised ML algorithms in resolving classification and clustering issues. Quantum ML aims to create more complex quantum algorithms that could potentially supplant conventional ML methods for specific problems. Algorithms such as QSVM, quantum K-nearest neighbour, and k-means clustering exhibit quantum speedups in distance computation. The feature space is converted into the quantum feature space by the second-order expansion quantum feature mapping method.

In their work, J. Biamonte et al. [25] elucidated how the convergence of machine learning and quantum computing facilitated the development of algorithms for quantum machine learning. Machine learning, by conforming to quantum mechanical rules, enhanced the control and benchmarking of quantum systems. The reduction in computational complexity enhances the performance of quantum systems. They have discussed GSA for amplitude amplification, QSVM for classification tasks, k-means clustering for clustering operations, and numerous other quantum methodologies for big data processing. The authors noted that these approaches offer a quantum speedup.

M. Schuld et al. [26] discussed two methodologies for developing a quantum computing categorisation model. A quantum computer can approximate the inner product of its quantum states within feature Hilbert space to calculate a classical kernel, which can subsequently be utilised in any kernel method, such as Support Vector Machine (SVM), through the implicit approach. Secondly, there exists the explicit technique, which employs a variational quantum device to train a decision boundary utilising feature Hilbert space for classification.

N. Mishra et al. provided a review on QML, addressing the implications of quantum computing on machine learning and classical computing. We discuss the applications and techniques for executing quantum algorithms such as HHL, QSVM, and VQC. The authors briefly addressed data processing and visualisation techniques for QML-based models [27].

|e-ISSN: 2319-8753, p-ISSN: 2347-6710| www.ijirset.com | Impact Factor: 8.118|

|| Volume 11, Issue 9, September 2022 ||

| DOI:10.15680/IJIRSET.2022.1109004 |

V. Havlicek et al. [28] proposed and implemented a quantum variational classifier and a quantum kernel estimator. The authors addressed the issue of how quantum variational classifiers and quantum kernel estimators manage classification challenges. The former employs a variational circuit to address the issues, whereas the latter computes the kernel matrix. Employing quantum state-space as the feature space is essential for the advancement of both algorithms. This domain can be effectively accessed using quantum computers, offering a quantum advantage. Binary classification problems in QML are addressed utilising both methodologies.

S. Ahmed [29] built a QSVM-based classification model using an image dataset, which is difficult to implement on conventional computers due to the extensive number of features it necessitates. The authors discovered that the classification model exhibits inferior performance on a quantum computer owing to the intrinsic noise present in quantum systems.

Y. Suzuki et al. [30] proposed a methodology for selecting an optimal feature map for QSVM. The authors provided a formula to determine the minimum training accuracy, indicating if the selected feature map is effective for linearly separating datasets. A synthesis approach is also offered that use a mixture of kernels to generate a more precise feature map across a broader spectrum of features.

In the classification of the breast cancer dataset, S. Saini et al. [31] proposed a QSVM-based model and conducted a comparison with SVM and VQC-based alternatives. The intricate calculations performed on the quantum simulator revealed that the QSVM-based model was inferior to the traditional SVM regarding accuracy. The QSVM model surpasses the SVM model by a factor of 234 in speed, demonstrating a quantum advantage.

III. QUANTUM MACHINE LEARNING TECHNIQUES

Quantum Machine Learning (QML) is an emerging discipline that combines machine learning with quantum computing to analyze data using quantum computers. The superposition concept is used by quantum computers and simulators to store data in several states, which allows for faster processing by performing actions on these states simultaneously [27]. Algorithms provided by QML are designed to tackle complex problems that traditional ML struggles with. In order to run on quantum computers, conventional algorithms are transformed into quantum algorithms. Deep neural networks and other traditional machine learning technologies excel in discovering statistical patterns in data and then creating new data that mirrors those patterns. There are statistical patterns in quantum systems that are difficult for classical computers to reproduce, but quantum algorithms may be able to spot these patterns more effectively than conventional algorithms. Quantum ML methods include supervised and unsupervised learning techniques. Quantum algorithms achieve their goals by encoding data onto qubits, running it via unitary operations, and then measuring the qubits' states.

Working of Classical machine learning and quantum machine learning are shown in figure 1.

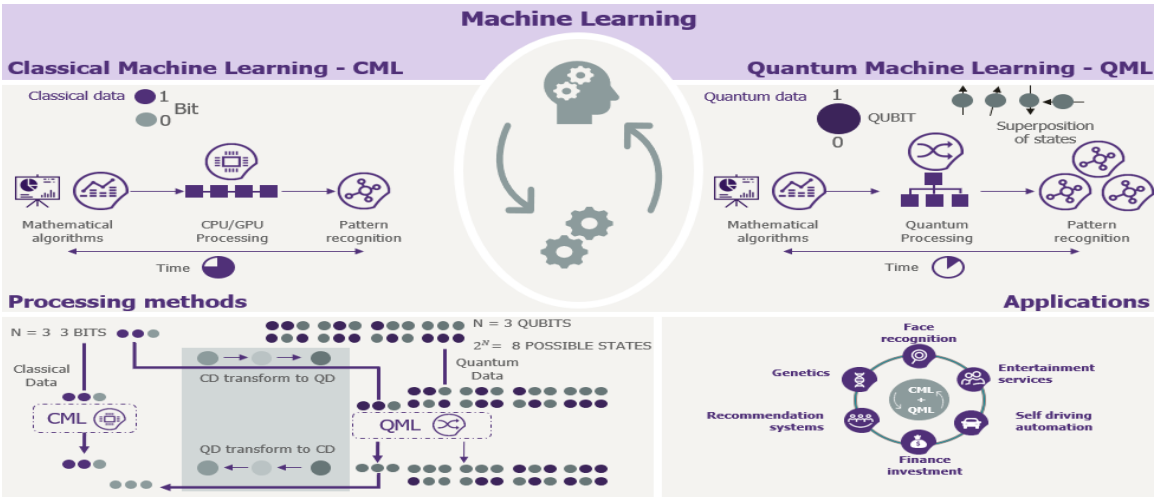

Figure 1: Working of CML and QML

|e-ISSN: 2319-8753, p-ISSN: 2347-6710| www.ijirset.com | Impact Factor: 8.118|

|| Volume 11, Issue 9, September 2022 ||

| DOI:10.15680/IJIRSET.2022.1109004 |

Quantum Support Vector Machine

Machine learning encompasses several data classification techniques applicable across diverse fields, including bioinformatics, picture classification, biological research, and facial recognition. The utilisation of algorithms in these methods enables computers to identify and classify data into distinct categories. Classical machine learning extensively use the renowned support vector machine (SVM) classification approach, whereas quantum machine learning utilises the QSVM algorithm [22]. Support Vector Machines employ kernel methods to address non-linear classification problems. However, there are circumstances in which addressing multiple dimensions is essential for resolving an issue. The augmentation of the sparse matrix depicting the data points introduces complexity and heightens the risk of overfitting, often referred to as the curse of dimensionality. In scenarios involving more dimensions than classical computers can manage, QSVM emerges as a compelling alternative. QSVM exhibits exponential acceleration in classification tasks by using the potential of quantum simulators and computers. Quantum algorithms in the QML domain facilitate the intentional misclassification of data, aiding in the mitigation of overfitting. The incorporation of the slack variable ξi, subject to the constraint $i \geq 0$, facilitates the measurement of misclassifications. To determine w and b, we must address the optimisation problem presented by the equation below.

$$(w, b) = \min_{w,b} \left[\frac{1}{2}\|w\| + C\sum_{i=1}^{N} \xi_i\right]$$

For experimental work, kaggle heart disease data set is used. The experiment was performed on IBMQ using Qiskit . QSVM and SVM were implemented in the IBMQ. The results are shown in figure 2 and figure 3.

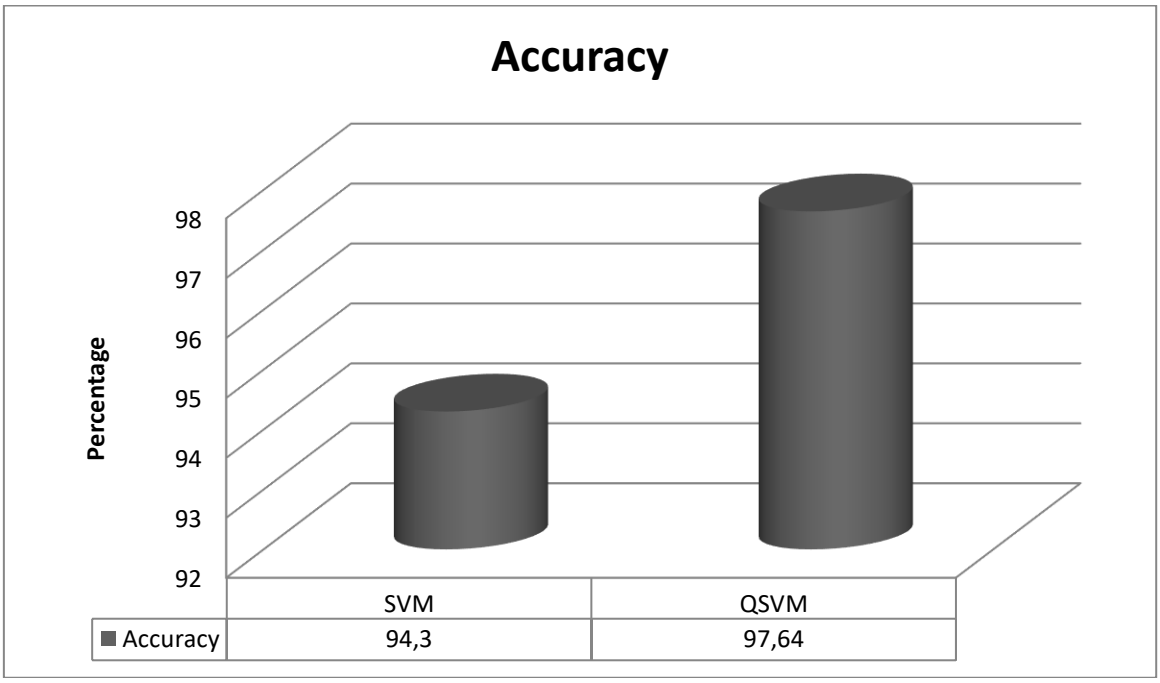

Figure 2: Accuracy of SVM and QSVM

|e-ISSN: 2319-8753, p-ISSN: 2347-6710| www.ijirset.com | Impact Factor: 8.118|

|| Volume 11, Issue 9, September 2022 ||

| DOI:10.15680/IJIRSET.2022.1109004 |

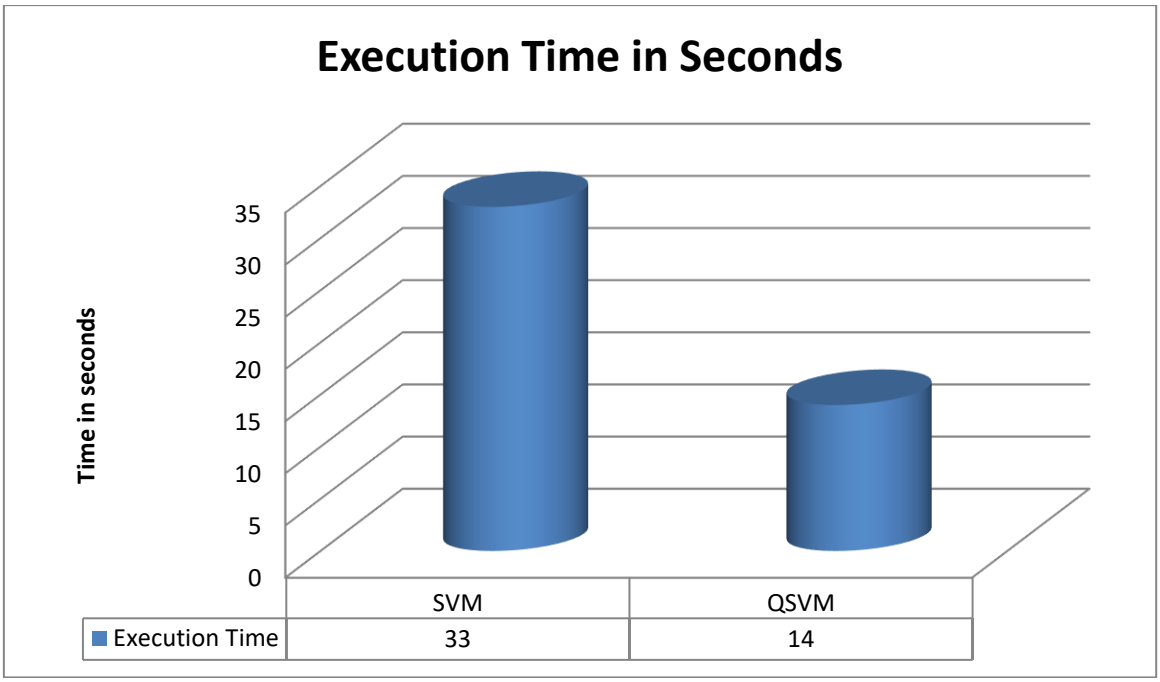

Figure 3: Execution Time of SVM and QSVM

IV. CONCLUSION

Quantum machine learning (QML) exploits the processing power of quantum systems to overcome the limitations of traditional machine learning approaches and therefore represents a new paradigm for data analysis. QML reveals the potential for rapid data processing with enhanced model accuracy and powerful analysis of massive datasets via quantum algorithms in supervised learning—namely, Quantum Support Vector Machines (QSVMs), Quantum Neural Networks (QNNs) and in unsupervised learning—specifically, Quantum Principal Component Analysis (QPCA) and Quantum K-Means Clustering.Although the current stage of Noisy Intermediate-Scale Quantum (NISQ) devices suffers from numerous challenges including noise, gate count and error correction, hybrid quantum-classical methods provide a realistic path toward harnessing quantum advantages. Since the application of quantum algorithms on actual data is possible with these techniques, they are helpful in many fields such as banking, healthcare, material science, and logistics.In the long run, we will be able to harness the potential of QML more fully over quantum technology advances, thus allowing breakthroughs in addressing hard-to-approach, high-dimensional problems. QML is becoming key to the advancements in AI and computational science by motivating further research in scalable quantum systems, error correcting codes and quantum algorithms. Hazardous that a this quantum edge would unlock new opportunities for innovation and creativity within the next era.

REFERENCES

[1] F. Bova, A. Goldfarb, and R. G. Melko, “Commercial applications of quantum computing,” EPJ Quantum Technology, vol. 8, no. 1, pp. 1–13, 2021, doi: 10.1140/epjqt/s40507-021-00091-1.

[2] L. Gyongyosi and S. Imre, “A Survey on quantum computing technology,” Computer Science Review, vol. 31, pp. 51–71, 2019, doi: 10.1016/j.cosrev.2018.11.002.

[3] A. Giani and Z. Eldredge, “Quantum Computing Opportunities in Renewable Energy,” SN Computer Science, vol. 2, no. 393, 2021.

[4] J. Roffe, “Quantum error correction: an introductory guide,” Contemporary Physics, vol. 60, no. 3, pp. 226–245, 2019, doi: 10.1080/00107514.2019.1667078

[5] A. Mandviwalla, K. Ohshiro, and B. Ji, “Implementing Grover’s Algorithm on the IBM Quantum Computers,” Proceedings - 2018 IEEE International Conference on Big Data, Big Data 2018, pp. 2531–2537, 2019, doi: 10.1109/BigData.2018.8622457.

|e-ISSN: 2319-8753, p-ISSN: 2347-6710| www.ijirset.com | Impact Factor: 8.118|

|| Volume 11, Issue 9, September 2022 ||

| DOI:10.15680/IJIRSET.2022.1109004 |

[6] A. J. et al., "Quantum Algorithm Implementations for Beginners," ACM Transactions on Quantum Computing, vol. 3, no. 4, pp. 1–92, 2022, doi: 10.1145/3517340.
[7] M. Usama et al., "Unsupervised Machine Learning for Networking: Techniques, Applications and Research Challenges," IEEE Access, vol. 7, pp. 65579–65615, 2019, doi: 10.1109/ACCESS.2019.2916648.
[8] P. Rebentrost, M. Mohseni, and S. Lloyd, "Quantum support vector machine for big data classification," Physical Review Letters, vol. 113, no. 3, pp. 1–5, 2014, doi: 10.1103/PhysRevLett.113.130503.
[9] Sent´ıs, G., Calsamiglia, J., Munoz-Tapia, R., & Bagan, E. (2012). Quantum learning without quantum memory. Scientific reports, 2 , 1–8. doi:10.1038/srep00708.
[10] Lloyd, S., Mohseni, M., & Rebentrost, P. (2013). Quantum algorithms for supervised andunsupervised machine learning. arXiv:1307.0411v2 [quant-ph], . URL: https://arxiv.org/pdf/1307.0411.pdf.
[11] Rebentrost, P., Mohseni, M., & Lloyd, S. (2014). Quantum support vector machine for bigdata classification. Physical review letters, 113 , 130503. doi:10.1103/PhysRevLett.113.130503.
[12] Schuld, M., Sinayskiy, I., Petruccione, F., & Pham, D.-N. (2014). Quantum computingfor pattern classification. In D.-N. Pham, & S.-B. Park (Eds.), PRICAI 2014: Trendsin Artificial Intelligence. Pacific Rim International Conference on Artificial Intelligence(PRICAI) 2014. Lecture Notes in Computer Science (pp. 208–220). Gold Coast, QLD,Australia: Springer, Cham volume 8862. doi:10.1007/978-3-319-13560-1_17.
[13] Wiebe, N., Kapoor, A., & Svore, K. M. (2015). Quantum nearest-neighbor algorithms for machine learning. Quantum Information and Computation, 15 , 318–358. URL: https://www.microsoft.com/en-us/research/publication/quantum-nearest-neighbor-algorithms-for-machine-learning/.
[14] Farhi, E., & Neven, H. (2018). Classification with quantum neural networks on near term processors. arXiv:1802.06002v2 [quant-ph], . URL: https://arxiv.org/pdf/1802. 06002.pdf.
[15] Kerenidis, I., & Luongo, A. (2018). Quantum classification of the mnist dataset via slowfeature analysis. arXiv:1805.08837v2[quant-ph], . URL: https://arxiv.org/pdf/1805. 08837.pdf.
[16] Torrontegui, E., & Garc´ıa-Ripoll, J. J. (2019). Unitary quantum perceptron as efficient universal approximator. EPL (Europhysics Letters), 125 , 30004. doi:10.1209/ 0295-5075/125/30004.
[17] Liang, J.-M., Shen, S.-Q., Li, M., & Li, L. (2020). Variational quantum algorithms for dimensionality reduction and classification. Physical Review A, 101 , 032323. doi:10. 1103/PhysRevA.101.032323.
[18] Sergioli, G. (2020). Quantum and quantum-like machine learning: A note on differences and similarities. Soft Computing, 24 , 10247–10255. doi:10.1007/s00500-019-04429-x.
[19] Pande, M., & Mulay, P. (2020). Bibliometric survey of quantum machine learning. Science & Technology Libraries, (pp. 1–14). doi:10.1080/0194262X.2020.1776193.
[20] Abohashima, Z., Elhosen, M., Houssein, E. H., & Mohamed, W. M. (2020). Classification with quantum machine learning: A survey. arXiv:2006.12270v1 [quant-ph], . URL: https://arxiv.org/ftp/arxiv/papers/2006/2006.12270.pdf.
[21] S. Lloyd, M. Mohseni, and P. Rebentrost, "Quantum algorithms for supervised and unsupervised machine learning," pp. 1–11, 2013, [Online]. Available: http://arxiv.org/abs/1307.0411
[22] P. Rebentrost, M. Mohseni, and S. Lloyd, "Quantum support vector machine for big data classification," Physical Review Letters, vol. 113, no. 3, pp. 1–5, 2014, doi: 10.1103/PhysRevLett.113.130503.
[23] J. Adcock et al., "Advances in quantum machine learning," 2015, [Online]. Available: http://arxiv.org/abs/1512.02900
[24] M. Schuld, I. Sinayskiy, and F. Petruccione, "An introduction to quantum machine learning," Contemporary Physics, vol. 56, no. 2, pp. 172–185, 2015, doi: 10.1080/00107514.2014.964942.
[25] J. Biamonte, P. Wittek, N. Pancotti, P. Rebentrost, N. Wiebe, and S. Lloyd, "Quantum machine learning," Nature, vol. 549, no. 7671, pp. 195–202, 2017, doi: 10.1038/nature23474.
[26] M. Schuld and N. Killoran, "Quantum Machine Learning in Feature Hilbert Spaces," Physical Review Letters, vol. 122, no. 4, 2019, doi: 10.1103/PhysRevLett.122.040504.
[27] N. Mishra et al., "Quantum Machine Learning: A Review and Current Status," Advances in Intelligent Systems and Computing, vol. 1175, no. September, pp. 101–145, 2021, doi: 10.1007/978-981-15-5619-7_8.
[28] V. Havlíček et al., "Supervised learning with quantum-enhanced feature spaces," Nature, vol. 567, no. 7747, pp. 209–212, 2019, doi: 10.1038/s41586-019-0980-2.
[29] S. Ahmed, "Pattern Recognition with Quantum Support Vector Machine(QSVM) on Near Term Quantum Processors," Brac University, 2019.
[30] Y. Suzuki et al., "Analysis and synthesis of feature map for kernel-based quantum classifier," Quantum Machine Intelligence, vol. 2, pp. 1–9, 2020, doi: 10.1007/s42484-020-00020-y
[31] S. Saini, P. Khosla, M. Kaur, and G. Singh, "Quantum Driven Machine Learning," International Journal of Theoretical Physics, vol. 59, no. 12, pp. 4013–4024, 2020, doi: 10.1007/s10773-020-04656-1.

INTERNATIONAL STANDARD SERIAL NUMBER INDIA

INNO SPACE
SJIF Scientific Journal Impact Factor

doi cross ref

निस्केयर
NISCAIR

INTERNATIONAL JOURNAL OF INNOVATIVE RESEARCH

IN SCIENCE | ENGINEERING | TECHNOLOGY

9940 572 462 6381 907 438 ijirset@gmail.com

www.ijirset.com

Scan to save the contact details